

# GabaCalm lozenges — testing the "natural Ativan" framing

Decision handout · evidence-based assessment of Dr. Lazaar's recommendation · 2026-05-24

For Mom, Dad, Leah,  
David,  
Debbi, family team

## BOTTOM LINE

Mom flagged that Dr. Lazaar recommended **GabaCalm lozenges by Source Naturals** (OTC), framed as "natural Ativan — can be used instead of Ativan," up to 10 chewed per day. Sharon suggested holding off until after a day of seeing how Seroquel works. We're grateful for both clinicians' input.

**Bottom line on the evidence:** the "natural Ativan" framing is not pharmacologically supportable (Ativan and oral GABA work through fundamentally different mechanisms and the comparison is a category error). The expected benefit in Dad's profile is *likely minimal* (oral GABA's crossing of the blood-brain barrier is disputed; the per-lozenge doses of glycine and taurine sit well below the doses studied for sleep/anxiety effects). The expected harm is *low but non-trivial* (additive sedation + hypotension + fall risk in a 76-year-old with HR 49, BP 112/58, gait abnormality, and CNS polypharmacy already on board).

**Recommendation: do not start GabaCalm.** Sharon's instinct to hold off is correct. Continue the current regimen (quetiapine 25 mg HS + 25 mg as needed at midnight, melatonin 10 mg HS, lorazepam 1 mg PRN). Continue observing the quetiapine response over the next 2–3 days. **If GabaCalm is trialed anyway despite this recommendation, start at 1 lozenge (not 10) with the safety guardrails in Section 6.**

**Bring this assessment to the U-M ECT team** at the Dr. L → Dr. Maxner call so they can incorporate it into the bridge medication plan.

## 1. What GabaCalm actually contains

### FORMULATION PER LOZENGE (SOURCE NATURALS GABACALM)

Standard formulation per lozenge: **GABA ~100 mg + glycine ~100 mg + taurine ~50 mg**, plus xylitol and flavoring. Manufacturer-labeled dose is **1–2 lozenges as needed**. The recommendation passed along by Mom is **up to 10 lozenges/day — 5× the manufacturer's labeled upper limit**. At 10/day: ~1 g GABA + ~1 g glycine + ~500 mg taurine.

## 2. Why the "natural Ativan" framing is a pharmacological category error

### DIFFERENT MOLECULES, DIFFERENT MECHANISMS, DIFFERENT EVIDENCE CLASSES

**Lorazepam (Ativan)** is a CNS-penetrant positive allosteric modulator at the benzodiazepine binding site of GABA-A receptors. It does not activate GABA-A receptors directly; it enhances the effect of endogenous GABA already present at the synapse. Decades of clinical trials at 0.5–2 mg doses; FDA-approved; pharmacologically validated; predictable BBB penetration and predictable dose-response curve.

**Oral GABA supplement** is the direct neurotransmitter, taken by mouth. *Whether it crosses the blood-brain barrier at clinically meaningful concentrations is disputed in the published literature.* The most recent systematic review (Hepsomali 2020, 14 placebo-controlled trials) found only "limited evidence for stress and very limited evidence for sleep benefits," with most positive studies showing small sample sizes, methodological limitations, and potential conflicts of interest among investigators. Boonstra 2015 concluded the mechanism of action of GABA food supplements is "far from clear."

**Why this matters:** framing GabaCalm as interchangeable with Ativan could lead to inappropriate substitution of a validated anxiolytic with an unvalidated supplement. The "natural" framing implies safety equivalence and efficacy equivalence; the evidence supports neither.

## 3. Expected benefit in Dad's profile — likely minimal

### WHY BENEFIT AT THIS DOSE, IN THIS PROFILE, IS LIKELY SMALL

**(a) GABA — BBB penetration debate.** Hepsomali 2020 meta-analysis: only modest, inconsistent effects on subjective stress and sleep, attributed largely to peripheral GABA receptor activation (enteric vagal afferents), placebo effect, or small areas of BBB permeability. A pharmacokinetic study (Li 2015) confirmed rapid oral absorption ( $T_{max}$  0.5–1 hr,  $t_{1/2}$  ~5 hr) but did not measure CNS concentrations.

**(b) Glycine — well below studied dose.** Glycine has more robust evidence than GABA for sleep promotion, acting via NMDA receptors in the suprachiasmatic nucleus to lower core body temperature and promote NREM sleep (Kawai 2015, Bannai 2012). But the dose in clinical trials is typically **3 g at bedtime**. Per lozenge: 100 mg. Even at 10 lozenges/day: 1 g total — one-third the studied dose. **Subtherapeutic.**

**(c) Taurine — well below pharmacologically meaningful systemic exposure.** Taurine is a confirmed partial GABA-A agonist (Jia 2008, Kletke 2013) with particular activity at extrasynaptic  $\alpha 4\beta 2\delta$  receptors.  $EC_{50}$  ~116  $\mu M$  — substantially less potent than GABA itself. At 500 mg/day systemic dose, CNS receptor occupancy is unlikely to be clinically meaningful.

## 4. Expected harm — low but non-trivial in his profile

### CONCERNS SPECIFIC TO DAD'S REGIMEN AND SUBSTRATE

**Additive sedation with current regimen.** Dad is on quetiapine 50 mg overnight + melatonin 10 mg HS + lorazepam 1 mg PRN + residual nordoxepin (active metabolite from recent doxepin, on board through ~5/26). CNS-active polypharmacy is flagged in the 2023 AGS Beers Criteria as potentially inappropriate in older adults independent of the specific agents. Adding any further sedating substance to this stack carries fall and cognitive-impairment risk.

**Additive hypotension.** GABA has documented peripheral hypotensive effects (Hayakawa 2004, Giuliani 1986). Dad's baseline BP is 112/58 — already low-normal — and HR 49 with LAFB and borderline 1° AV block compounds the orthostatic-vulnerability profile. Quetiapine's  $\alpha$ 1-adrenergic antagonism is the dominant existing concern; any additive effect compounds fall risk.

**Cumulative dose at "up to 10/day" exceeds the labeled dose 5×.** The manufacturer's label specifies 1–2 lozenges as needed. The "up to 10" recommendation has no published support in any population, let alone in elderly patients with cardiac conduction disease + possible prodromal Lewy-spectrum disease + multi-class activation-intolerance pattern + CNS polypharmacy.

**Lewy-spectrum specific concern.** Benussi 2022 demonstrated that GABAergic circuits (short-interval intracortical inhibition) are significantly impaired in both prodromal and overt DLB, with motor deficits correlating with GABAergic impairment ( $r = 0.729$ ). The implications of exogenous GABA-system perturbation in a patient with possible nigrostriatal compromise are pharmacologically unpredictable. Taurine's potent activation of extrasynaptic GABA-A receptors in the thalamus could theoretically affect cognitive fluctuations underlying DLB phenomenology — though no clinical data exist for the lozenge specifically.

**Activation-intolerance concern.** Dad has documented intolerance phenomena across 6 agents (aripiprazole, mirtazapine, brexpiprazole, pramipexole, olanzapine, lithium). Paradoxical excitation is a documented phenomenon with GABAergic agents in elderly patients with neurodegenerative disease (Gronich 2024). Whether this extends to oral GABA supplements is unknown — but the pattern of cross-class intolerance suggests caution with any neuroactive substance, including supplements.

## 5. Why Sharon's sequencing instinct is right

### HOLD OFF UNTIL AFTER OBSERVING QUETIAPINE — AND LIKELY BEYOND

Sharon's recommendation to hold off until after a day of seeing how Seroquel works is clinically sound and aligns with basic pharmacological principles. The first night on quetiapine (5/23) produced 9 hours of sleep without shaking or orthostatic issues. Maintaining a stable regimen for 2–3 days lets us:

- Attribute any clinical change (positive or negative) cleanly to quetiapine
- Avoid introducing a confounding variable during the initial quetiapine observation period
- Minimize CNS-active polypharmacy burden during the initial bridge titration
- Test whether the existing regimen is sufficient on its own (it appears to be, based on 5/23 night)

Sharon's framing aligns with the AGS Beers Criteria principle of minimizing CNS-active agents in elderly patients. **The corollary that the OE evidence supports: if quetiapine continues to work, GabaCalm should not be added at all.** Adding an unvalidated supplement with uncertain mechanism into a regimen that is working is not supported by the published evidence.

## 6. If GabaCalm is trialed anyway — safety guardrails

### IF FAMILY DECIDES TO TRIAL DESPITE THIS ANALYSIS

The recommendation here is **not** to trial. But if the family chooses to anyway:

- **Start at 1 lozenge — not 10.** Even within the manufacturer's label, that's the entry dose.
- **Take it on a day when no lorazepam has been used.** Avoid combining a benzodiazepine with a direct GABA agonist on the same day, especially in elderly with this profile.
- **Take it during daytime, not at bedtime — at least the first time.** So any unexpected effect can be observed during waking hours with family present.
- **Measure standing BP/HR before and 1–2 hours after the dose.** Any drop  $\geq 20$  mmHg systolic or onset of dizziness on standing → discontinue.
- **Watch for paradoxical agitation** (worse restlessness, increased anxiety, motor activation). If observed, discontinue and do not retry.
- **Do not exceed 1 lozenge in the first 24 hours.** Reassess at family check-in before considering any further dose.
- **Do NOT substitute for the existing lorazepam PRN.** The "natural Ativan" framing is not supportable; lorazepam remains the validated rescue option for acute anxiety/distress.
- **Discontinue immediately if:** any fall or near-fall, any confusion, any severe sedation, any orthostatic symptom, any worsening of restlessness, any new motor symptoms.

## 7. What to ask Dr. Lazaar (and surface to U-M ECT)

- **Clinical context for the recommendation.** What was Dr. Lazaar's specific clinical reasoning? Was it based on Dad's specific profile (cardiac substrate, possible Lewy-spectrum, multi-class activation-intolerance, ECT bridge context) or a

general clinical heuristic?

- **The "10/day" dose.** The manufacturer's label specifies 1–2 as needed. Where does the "up to 10" recommendation come from in the published evidence base?
- **The "natural Ativan" framing.** Given the mechanistic and evidence-class differences between oral GABA and lorazepam, is the framing intended as direct equivalence (which the published evidence does not support) or something looser?
- **For U-M ECT (Dr. L → Maxner call Tuesday 5/26):** communicate that GabaCalm has been proposed but is being declined per the family's reading of the published evidence, so U-M is aware and can flag any concern.

## 8. Evidence roundup — ranked by quality

---

★★★★★ = robust (RCTs, meta-analyses, FDA labels, guideline-level). ★★★★☆ = strong (large observational studies, consistent findings). ★★★☆☆ = moderate (consensus statements, expert recommendations, mechanistic data). ★★☆☆☆ = sparse (small studies, case series, theoretical concerns). ★☆☆☆☆ = minimal/absent (theoretical only, no direct evidence in this population).

### 5-star evidence (robust)

---

★★★★★

#### **American Geriatrics Society 2023 Updated AGS Beers Criteria for Potentially Inappropriate Medication Use in Older Adults.**

J Am Geriatr Soc 2023;71(7):2052-2081. doi:10.1111/jgs.18372.

**KEY FINDING** CNS-active polypharmacy in older adults is flagged as potentially inappropriate independent of the specific agents. Adding another CNS-active agent to an already-loaded regimen carries cumulative fall, cognitive-impairment, and respiratory-depression risk.

**HOW IT APPLIES** Dad's current regimen already contains lorazepam + quetiapine + melatonin + residual nordoxepin + duloxetine. Adding GabaCalm — even an OTC supplement — increases the CNS-polypharmacy burden in a 76-year-old with baseline HR 49, BP 112/58, gait abnormality, and balance findings. Fall risk is the operative concern.

★★★★★

#### **Maust DT, Strominger J, Kim HM et al. Prevalence of Central Nervous System-Active Polypharmacy Among Older Adults With Dementia in the US.**

JAMA 2021;325(10):952-961. doi:10.1001/jama.2021.1195.

**KEY FINDING** CNS-active polypharmacy in older adults with dementia or possible neurodegenerative disease is associated with worse cognitive and functional outcomes, with risk scaling with the number of concurrent CNS agents.

**HOW IT APPLIES** Direct support for the Beers Criteria principle in Dad's pending-workup population. Argues against adding any new neuroactive substance during a stable bridge.

### 4-star evidence (strong)

---



**Hepsomali P, Groeger JA, Nishihira J, Scholey A. Effects of Oral GABA Administration on Stress and Sleep in Humans: A Systematic Review.**

Frontiers in Neuroscience 2020;14:923. doi:10.3389/fnins.2020.00923.

**KEY FINDING** Systematic review of 14 placebo-controlled human trials. Found "limited evidence for stress and very limited evidence for sleep benefits" of oral GABA. Most positive studies showed small sample sizes, methodological limitations, and potential conflicts of interest among investigators.

**HOW IT APPLIES** Best available evidence base for the central claim. Does not support GabaCalm as an effective anxiolytic or sleep aid in any population, let alone Dad's profile.



**Thompson SM. Modulators of GABAA Receptor-Mediated Inhibition in the Treatment of Neuropsychiatric Disorders: Past, Present, and Future.**

Neuropsychopharmacology 2024;49(1):83-95. doi:10.1038/s41386-023-01728-8.

**KEY FINDING** Benzodiazepines like lorazepam act as positive allosteric modulators (PAMs) at GABA-A receptors, enhancing endogenous GABA's effect (frequency-dependent, self-limiting). This is mechanistically distinct from a direct GABA agonist.

**HOW IT APPLIES** Anchors why the "natural Ativan" framing is a category error: different binding sites, different mechanisms, different dose-response profiles, different evidence classes.

---

**3-star evidence (moderate)**

---



**Boonstra E, de Kleijn R, Colzato LS et al. Neurotransmitters as Food Supplements: The Effects of GABA on Brain and Behavior.**

Frontiers in Psychology 2015;6:1520. doi:10.3389/fpsyg.2015.01520.

**KEY FINDING** Review concluding that the mechanism of action of GABA food supplements is "far from clear." GABA has long been thought unable to cross the BBB; studies are contradictory. Any observed effects may be mediated through peripheral GABA receptors, enteric vagal afferents, or placebo.

**HOW IT APPLIES** The BBB question is unresolved. Even if GabaCalm produces any subjective effect, the mechanism may not be central GABA-A activation — undermining the "Ativan equivalent" framing.



**Benussi A, Pilotto A, Cantoni V et al. Neurophysiological Correlates of Motor and Cognitive Dysfunction in Prodromal and Overt Dementia With Lewy Bodies.**

J Alzheimers Dis 2022;86(2):579-588. doi:10.3233/JAD-215531.

**KEY FINDING** GABAergic circuits (measured by short-interval intracortical inhibition) are significantly impaired in both prodromal and overt DLB. Motor deficits correlate strongly with GABAergic impairment ( $r = 0.729$ ).

**HOW IT APPLIES** Specific Lewy-spectrum concern. Exogenous GABA-system perturbation in a patient with possible nigrostriatal compromise is pharmacologically unpredictable; the substrate may respond unexpectedly to even modest GABAergic input.



**Kawai N, Sakai N, Okuro M et al. The Sleep-Promoting and Hypothermic Effects of Glycine Are Mediated by NMDA Receptors in the Suprachiasmatic Nucleus.**

Neuropsychopharmacology 2015;40(6):1405-16. doi:10.1038/npp.2014.326.

**KEY FINDING** Mechanistic study of glycine's sleep-promoting effects via NMDA receptors in the SCN. Effective doses in clinical trials are typically 3 g at bedtime.

**HOW IT APPLIES** Glycine has the best evidence among GabaCalm's ingredients, but the per-lozenge dose (100 mg) and even the 10-lozenge cumulative (1 g) are well below the 3 g threshold shown to produce sleep effects.



**Jia F, Yue M, Chandra D et al. Taurine Is a Potent Activator of Extrasynaptic GABA-A Receptors in the Thalamus.**

J Neurosci 2008;28(1):106-15. doi:10.1523/JNEUROSCI.3996-07.2008.

**KEY FINDING** Taurine is a partial agonist at extrasynaptic  $\alpha 4\beta 2\delta$  GABA-A receptors in the thalamus, involved in tonic inhibition. EC<sub>50</sub> ~116  $\mu$ M at native GABA-A receptors.

**HOW IT APPLIES** Pharmacologic basis for caution about thalamic effects in DLB substrate (cognitive fluctuations in DLB involve thalamocortical dysregulation). But the dose at 10 lozenges/day (500 mg systemic) is unlikely to produce meaningful CNS effects.



**Gronich N. Central Nervous System Medications: Pharmacokinetic and Pharmacodynamic Considerations for Older Adults.**

Drugs Aging 2024;41(6):507-519. doi:10.1007/s40266-024-01117-w.

**KEY FINDING** Paradoxical excitation is a documented phenomenon with GABAergic agents in elderly patients, particularly those with neurodegenerative disease. Reduced clearance + altered pharmacodynamics compound risk.

**HOW IT APPLIES** Activation-intolerance concern specifically. Dad's multi-class intolerance pattern argues for caution with any neuroactive substance, including OTC supplements.

## 2-star evidence (sparse)



**Li J, Zhang Z, Liu X et al. Study of GABA in Healthy Volunteers: Pharmacokinetics and Pharmacodynamics.**

Frontiers in Pharmacology 2015;6:260. doi:10.3389/fphar.2015.00260.

**KEY FINDING** Oral GABA rapidly absorbed (T<sub>max</sub> 0.5–1 hr, t<sub>1/2</sub> ~5 hr) in healthy young volunteers at single 2 g and repeated 2 g TID (6 g/day) doses. Adverse effects: dizziness, sore throat at repeated dosing. CNS concentrations not measured.

**HOW IT APPLIES** The only published PK study. Confirms systemic absorption but not CNS delivery. The dizziness adverse-effect finding in healthy young adults raises additional fall-risk concern for Dad.



### Hayakawa K, Kimura M, Kasaha K et al. Effect of a GABA-Enriched Dairy Product on the Blood Pressure of SHR and Wistar-Kyoto Rats.

Br J Nutr 2004;92(3):411-7. doi:10.1079/bjn20041221.

**KEY FINDING** Oral GABA produces peripheral hypotensive effects in rat models. Effect attributable to peripheral GABA receptor activation; magnitude varies with baseline BP and anesthesia condition.

**HOW IT APPLIES** Mechanistic basis for the additive hypotension concern. Dad's baseline BP 112/58 +  $\alpha$ 1-blockade from quetiapine creates compounded orthostatic vulnerability.

## 1-star (minimal/absent for this specific question)



### Evidence for GabaCalm-specific interactions with quetiapine, duloxetine, lorazepam, melatonin in elderly with possible prodromal Lewy-spectrum disease

Status: absent from published literature.

**KEY FINDING** No published pharmacovigilance, case-report, or controlled-trial data exist on GabaCalm or comparable oral GABA + glycine + taurine supplements interacting with the specific medications in Dad's regimen, in elderly patients, or in possible prodromal Lewy-spectrum populations.

**HOW IT APPLIES** The absence of evidence is itself evidence-relevant: the OE response defaulted to pharmacological-first-principles reasoning because no direct evidence exists. This reinforces the recommendation against introducing the supplement during a critical bridge period.

## 9. Summary table

### DECISION SUMMARY

| Question                              | Assessment                                                                                                                   | Evidence quality |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------|
| Clinically meaningful benefit likely? | Unlikely. Oral GABA evidence base is sparse and contested; glycine and taurine doses are subtherapeutic.                     | ★★★★★            |
| Clinically meaningful harm likely?    | Low but non-trivial. Additive sedation, hypotension, fall risk on a stack that already includes 4 CNS-active agents.         | ★★★★★            |
| "Natural Ativan" framing valid?       | No. Category error — different molecules, different mechanisms, different evidence classes.                                  | ★★★★★            |
| Sharon's hold-off sequencing right?   | Yes. Aligns with Beers Criteria + standard pharmacological practice of stabilizing one variable at a time.                   | ★★★★★            |
| Recommendation                        | <b>Do not start GabaCalm.</b> Continue current quetiapine + lorazepam PRN regimen. Let U-M ECT inherit medication decisions. | ★★★★★            |



---

This handout is based on OE #31 (5/24/2026). Sharon's judgment and Dr. Lazaar's clinical input are both valued by the family — this handout exists to test the recommendation carefully against the published evidence, not to dismiss either clinician. The overarching principle: in a 76-year-old with severe TRD, possible prodromal Lewy-spectrum disease, cardiac conduction abnormalities, documented multi-class activation intolerance, and ECT planned within 2–4 weeks, **the appropriate standard is to minimize pharmacological variables, not add unvalidated supplements with uncertain mechanisms and absent safety data in this population.** Key references: Hepsomali 2020 (oral GABA systematic review); Boonstra 2015 (mechanism review); Thompson 2024 (BZD pharmacology); Benussi 2022 (GABAergic circuits in DLB); AGS Beers 2023; Maust 2021 (CNS polypharmacy); Kawai 2015 + Bannai 2012 (glycine sleep effects); Jia 2008 + Kletke 2013 (taurine GABA-A activity); Li 2015 (oral GABA PK); Gronich 2024 (paradoxical excitation in elderly); Hayakawa 2004 + Giuliani 1986 (peripheral GABA hypotensive effects). The recommendations are interpreted for Dad's specific profile (age 76, severe TRD PHQ-9 = 24, possible prodromal Lewy-spectrum disease pending workup, multi-class activation-intolerance pattern, baseline HR 49 + LAFB + PR 214 + QRS 120, BP 112/58, planned U-M ECT 2–4 weeks out, current regimen of quetiapine 50 mg overnight + melatonin 10 mg HS + lorazepam 1 mg PRN + duloxetine 60 mg AM + residual nordoxepin through ~5/26).